# MALAIKA FATIMA

**Computer Science Student | Aspiring AI & Game Development**

+92 322 6898750 • auratech1101@gmail.com • Pakistan • github.com/auratech01 • linkedin.com/in/auratech01

## PROFESSIONAL SUMMARY

Computer Science undergraduate with hands-on experience building multi-threaded Python applications and interactive HTML5 Canvas-based web games. Comfortable combining practical software development with modern AI tools such as the Google Gemini API and real-time system monitoring libraries. Seeking an internship or entry-level role in software development, applied AI, or game development to contribute, learn, and grow as a developer.

## EDUCATION

**Bachelor of Science in Computer Science (BSCS)** 2023 – Present
*Virtual University of Pakistan*
Currently in 7th Semester | CGPA: 3.72
**Core Coursework:** Object-Oriented Programming (C++), Database Management Systems (SQL), Web Systems & Technologies

## TECHNICAL SKILLS

**Programming Languages:** Python, JavaScript (ES6+), C++, SQL, HTML5, CSS3
**Web & Game Development:** HTML5 Canvas (2D), Interactive Game Loops, 2D Physics & Collision Handling, Drag-and-Drop Mechanics, CSS 3D Transforms, DOM Manipulation, REST APIs
**Frameworks & Libraries:** psutil, pywebview, CustomTkinter, Tkinter, Google Gemini API, Multi-Threading (Python), SMTP/Email Automation
**Developer Tools:** VS Code, Git, GitHub Desktop, Cisco Packet Tracer, Linux (Ubuntu), Figma

## FEATURED PROJECTS

**Project Sentinel — Desktop Security Monitoring & AI Code-Auditing Application**
*Python, JavaScript (ES Modules), HTML5/CSS3, pywebview, psutil, Google Gemini API*

- Built a multi-threaded Python security monitor with real-time detectors for firewall status, sustained CPU/RAM overload, suspicious outbound connections, failed logins and internet loss, each triggering a severity-based email alert.
- Designed a secure local architecture: token-authenticated REST API on localhost, PBKDF2 password hashing, persistent login lockout, view-only guest mode and OTP-based password recovery.
- Developed an offline pattern-based code scanner with optional Google Gemini API analysis, and a network port auditor that explains why each connection is flagged.
- Rebuilt the interface as a responsive HTML/CSS/JS glassmorphism dashboard with live ECG-style health graphs, and wrote 117 automated unit tests.

**Jade Lantern — Interactive Restaurant Web Experience**
*HTML5, CSS3, Vanilla JavaScript*
Live Site: https://auratech01.github.io/jade-lantern-restaurant/

- Designed a fine-dining restaurant website with animated page sections and a custom 3D flip-book style interactive menu built using CSS 3D transforms.
- Implemented a dynamic, JavaScript-driven menu system with fully responsive layouts and no external UI frameworks.

**Birthday Experience — Interactive 22-Level Web Game** DESKTOP ONLY
*HTML5 Canvas, JavaScript (ES6+), CSS3*
Live Site: https://auratech01.github.io/birthday-experience/ (Desktop Only)

- Built a 22-level interactive escape-room-style web game (engineered exclusively for desktop displays) using vanilla JavaScript state management for level progression and screen transitions.
- Implemented custom canvas-based mechanics, including physics-driven drag-and-drop mini-games and real-time particle animations (fireworks, confetti).
- Deployed as a lightweight, dependency-free game directly to GitHub Pages.

## INDEPENDENT PROJECT EXPERIENCE

**Self-Directed Software Developer — Personal & Academic Projects** 2024 – Present

- Developed responsive web applications, canvas-based rendering systems, and Python desktop tools independently and as part of academic coursework.
- Applied object-oriented programming principles in C++ and Python to build modular, maintainable code across multiple personal projects.

## CERTIFICATIONS

- CCNA: Introduction to Networks — Cisco Networking Academy
- CPA: Programming Essentials in C++ — Cisco Networking Academy